Inapproximability of Maximum Hamiltonian Path-S

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Overview

In this project, we define the *Maximum Hamiltonian Path-*S *problem* ($\mathsf{MaxHamPathS}$) and provide an inapproximability result for it through a series of reductions starting from $\mathsf{3SAT}$.

Introduction

One of the fundamental results in computational complexity theory is the *Cook-Levin theorem* [1], which proves the existence of an NP-complete language (SAT). Its importance transcends computational complexity theory, starting the theory of NP-complete problems (where it was discovered that many problems in different fields were NP-hard) and allowed the formulation of one of the deepest mysteries of computation and mathematics: $\mathsf{P} \neq \mathsf{NP}$?

This generated a strong interest in the scientific community, which motivated to understand the complexity of calculating *exact* solutions to such problems. Since $\mathsf{P} \neq \mathsf{NP}$ would imply the impossibility of finding such solutions *efficiently*, there was more interest in finding *approximate* solutions, which was sufficient for many problems in practice. But except for a few problems, no approximation algorithms were found and a *hardness of approximation*[1] began to be conjectured for such problems.

The PCP theorem was, unsuspectedly, the starting point that allow us to determine hardness of approximation for many NP-hard problems, which was an achievement for the field and it significantly influenced the design of modern algorithms and protocols. It should be mentioned that this result arose from problems seemingly unrelated to hardness of approximation[2] and was originally formulated as an extraordinarily counter intuitive characterization of NP.

However, there are NP-hard problems for which we still don't know whether we can approximate them efficiently, or they are NP-hard to approximate (inside certain range of approximation). In this project, we consider a variation of the *Hamiltonian Path problem* (which was one of the first problems shown to be NP-complete shortly after the Cook-Levin theorem) and show a hardness of approximation result for it using one version of the PCP theorem: the celebrated *Håstad's 3-bit PCP theorem* [2].

[1] In other words, calculating approximate solutions is as difficult as calculating optimal solutions.

[2] For which the concept of *interactive proof system* was introduced.

Preliminaries

Recall that, in SAT, we are given a boolean formula ϕ given in *conjunctive normal form* (CNF), with n boolean variables $x_1, ..., x_n$, and m clauses, and we need to decide if there is an assignment of its variables such that the formula ϕ is satisfied. One variant of SAT, which is also NP-complete, is 3SAT, where each clause has exactly 3 literals. Also, recall that in a graph, a *Hamiltonian* path is a path that visits each node of the graph exactly once. Thus, the Hamiltonian path problem is defined as follows:

Definition (Hamiltonian path problem)
Instance: An unweighted directed graph $G = (V, E)$*, a start node, and a final node.*
Question: Does G *has a Hamiltonian path that starts in the start node and ends in the end node?*

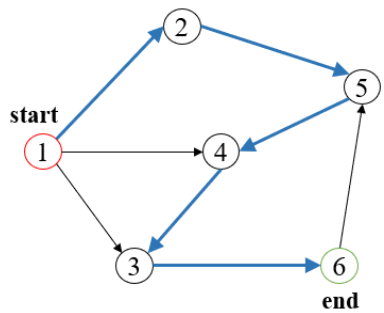

Figure 1: Hamiltonian path starting at node 1 and ending at node 6.

3SAT is very useful to determine NP-hardness for many problems. In fact, it can be used to prove that the Hamiltonian path problem is NP-complete. However, in order to have a hardness of approximation result, we need to define a natural *maximization* version of this problem:

Definition (Maximum Hamiltonian Path-S problem (MaxHamPathS))
Instance: An unweighted directed graph $G = (V, E)$*, and a subset* $S \subseteq V$ *such that there is a path that visits every node of* S *such that each node in the path is visited exactly once.*
Problem: Find one of such paths[3] *that maximizes its number of nodes.*

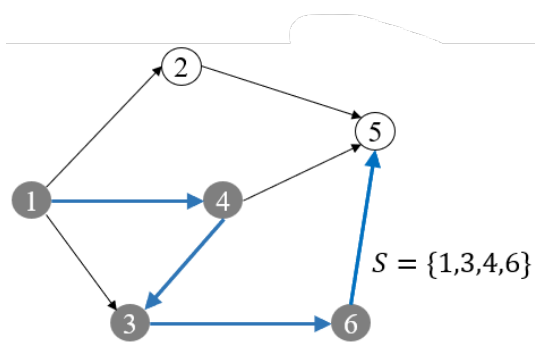

Figure 2: Hamiltonian path visiting all elements in S as well as node 5 not in S.

[3] I.e., a path that visits each node of S such that each node in the path is visited exactly once (which will be called an S-path).

(Figure 2 provides an example for this problem). Now, let A be a maximization problem and x an instance of A. In general, we will be interested in computing $\mathtt{OPT}(x)$. For any variation of SAT, $\mathtt{OPT}(x)$ is simply the maximum number of clauses satisfied. For MaxHamPathS, $\mathtt{OPT}(x)$ is the maximum number of nodes in the path. Similarly, we will be interested in computing $\mathtt{VAL}(x) := \mathtt{OPT}(x)/|x|$. I.e., for any variation of SAT, $\mathtt{VAL}(x)$ will simply be the maximum fraction of clauses satisfied. For MaxHamPathS, $\mathtt{VAL}(x)$ will be the maximum fraction of nodes in the path. Now, we will use the following problem:

Definition ($\mathsf{Gap}(A, a, b)$)
Let A be a maximization problem and $a, b : \mathsf{N} \to \mathsf{N}$ such that $b(n) < a(n)$ for all $\in \mathsf{N}$.
Input: An instance x of A such that $\mathtt{OPT}(x) \geq a(|x|)$ or $\mathtt{OPT}(x) \leq b(|x|)$.
Question: Determine which is the case.

We will use the following result, whose proof could be found in lemma 9.13 of [3], page 237.
Lemma 1. Suppose that $\mathsf{Gap}(A, a, b)$ is NP-hard. If there is a polynomial time algorithm such that for every input x, computes a number $\mathtt{ALG\text{-}OPT}(x)$ with

$$\mathtt{ALG\text{-}OPT}(x) > \frac{b(|x|)}{a(|x|)}\mathtt{OPT}(x), \quad \text{then } \mathsf{P} = \mathsf{NP}.$$

Reduction Proof Outline

First, we use a known gap amplification reduction from 3SAT to Max3SAT (Håstad PCP theorem); then, we give an L-reduction from Max3SAT to Max3SAT-7. Finally, we give an L-reduction from the Max3SAT-7 to MaxHamPathS as the main contribution of our project (see Figure 3).

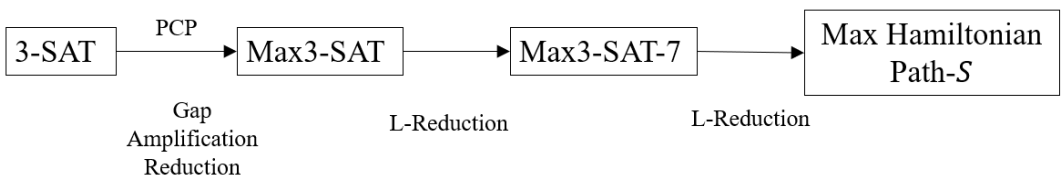

Figure 3: Reduction proof outline

Max3SAT Inapproximability Result

An interesting result obtained from straightening the PCP theorem was proved by Håstad in [2], which is an inapproximability result for the Max3SAT (known as the 3-bit PCP theorem, see theorem 9.34 of [3], page 246), which states the following:

Theorem 1 (Håstad PCP)
For every $\epsilon > 0$,[4] $\mathsf{Gap}(\mathsf{Max3SAT}, (\frac{7}{8} + \epsilon)m', (1 - \epsilon)m')$ is NP-hard, where m' is the number of clauses. I.e., there is a polynomial time reduction from 3SAT to Max3SAT $\phi \mapsto \phi'$ such that

$$\text{If } \phi \in \mathsf{3SAT}, \text{ then } \mathtt{VAL}(\phi') \geq 1 - \epsilon,$$
$$\text{If } \phi \notin \mathsf{3SAT}, \text{ then } \mathtt{VAL}(\phi') \leq \tfrac{7}{8} + \epsilon.$$

[4] Actually, $\epsilon \in (0, 1/16]$ in order to have $(\frac{7}{8} + \epsilon)m' < (1 - \epsilon)m'$.

Hence, using the previous lemma, $\mathsf{Max3SAT}$ is NP-hard to $(\frac{7}{8}+\delta)$-approximate for every $\delta > 0$.

Max3SAT-7 L-reduction from Max3SAT

Max3SAT-7 is a variant of Max3SAT where each variable occurs at most 7 times. We start with a Max3SAT formula $\phi^{'}$. We convert the Max3SAT formula $\phi^{'}$ into Max3SAT-7 formula $\phi^{''}$ by following steps:

1. If a variable x_i in formula $\phi^{'}$ occurs more than 7 times; there are two possible cases. If the number of repetitions is odd, we add a satisfying clause $(\overline{x_i} \vee x_i \vee \overline{x_i})$ bringing the number of repetitions of variable x_i to even. If the number of repetitions is even, we leave it as is.

2. Let k_i be the number of repetitions of variable x_i in formula $\phi^{'}$. We create k_i new variables $z_1, z_2, ..., z_{k_i}$. We replace these variables in place of x_i in the formula $\phi^{'}$.

3. We also create a 3-expander graph $G = (V, E)$ with $V = \{1, 2, .., k_i\}$ and edges such that each edge $(i, j) \in E$ has the clause $(z_i \vee \neg z_j) \wedge (z_j \vee \neg z_i)$ associated with it. The total number of edges and clauses in the 3-expander graph is $3k_i/2$ with each variable repeated

These 3 steps help convert the formula from $\phi^{'}$ to $\phi^{''}$ as it repeated for all variables, which occurs more than 7 times. The first step is to make sure that the number of repetitions greater than 7 is even. This is because there always exists a 3-expander graph for an even number of nodes which is crucial for this reduction.

The second step gets rid of variables whose repetitions are greater than 7 by replacing them with new variables, and those variables with less than 7 repetitions remain unchanged. The third and final step is the most important step, which forces all the new variables $z_1, z_2, ..., z_{k_i}$ created to replace variable x_i, will have the same truth value. This is because of the clause $(z_i \vee \neg z_j) \wedge (z_j \vee \neg z_i)$ which is only true if both truth values of z_i and z_j is either True or False.

One more important observation is that each new variable z_j occurs in the formula $\phi^{'}$ one time replacing the variable x_i and 2 times in each clause $(z_i \vee \neg z_j) \wedge (z_j \vee \neg z_i)$ and as it a 3-expander graph with the degree of each node is 3, it occurs 6 times in total in the graph. Hence, a total of exactly 7 times. Hence, a Max3SAT-7 formula.

But why is it an L-reduction? To prove this, we need to get familiar with two important terms, OPT and $\mathsf{BENEFIT}$. OPT is the maximum number of clauses satisfiable in a formula. $\mathsf{BENEFIT}$ is the number of clauses satisfied by a particular assignment. Let's determine these terms for Max3SAT and Max3SAT-7 and determine if there is a relation between them.

Let a and $\overline{a}$ an assignment for $\phi^{'}$ and $\phi^{''}$, respectively. Clearly,

$$\mathsf{OPT}(\phi^{''}) = \mathsf{OPT}(\phi^{'}) + 3 * \sum_{i=1}^{n'} k_i = \mathsf{OPT}(\phi^{'}) + 9m' = O(\mathsf{OPT}(\phi^{'}))$$

This is because there is $3*k_i$ additional number of clauses that are always satisfied in the 3-expander graph. The same can be said about the $\mathsf{BENEFIT}$.

$$\mathsf{BENEFIT}(\phi^{''}, \overline{a}) = \mathsf{BENEFIT}(\phi^{'}, a) + 3 * \sum_{i=1}^{n'} k_i = \mathsf{BENEFIT}(\phi^{'}) + 9m'$$

Hence, we have

$$|\mathsf{BENEFIT}(\phi^{'}, a) - \mathsf{OPT}(\phi^{'})| = O(|\mathsf{BENEFIT}(\phi^{''}, \overline{a}) - \mathsf{OPT}(\phi^{''})|)$$

Hence, this is a linear reduction from Max3SAT to Max3SAT-7 as it linearly preserves the gap. Observe that in the previous reduction, $n'' = \Theta(m')$. Consequently, we have the following result:

Theorem 2. Max3SAT $\leq_L$ Max3SAT-7. Moreover, this reduction $\phi' \mapsto \phi''$ satisfies

$$\mathtt{OPT}(\phi'') = \mathtt{OPT}(\phi') + 9m',$$

where m' is the number of clauses of ϕ'.

Max Hamiltonian Path-S L-reduction from Max3SAT-7

First, we will consider a reduction from 3SAT to Hamiltonian Path and we'll see what changes for our problem. We start with 3-CNF formula with 4 variables with $k = 2$ clauses given by:

$$\phi = (\overline{x_1} \vee x_2 \vee \overline{x_3}) \wedge (\overline{x_2} \vee x_3 \vee \overline{x_4})$$

We start constructing unweighted undirected graph from ϕ. We construct 4 paths $P_1, ..., P_4$ corresponding to 4 variables and each paths have $2 * k$ nodes connected to each other as shown in Figure 4(a). Then paths are interconnected to each other and the start and end node is added to the top and bottom of paths as shown in Figure 4(b). Finally, clause nodes C_1 and C_2 relating to two clauses in ϕ is connected to paths based on how they appear in the clause. If a variable x_i appears as x_i in clause node C_j, then the connection goes from path P_i to clause node and then from clause node to path P_i. If a variable x_i appears as $\overline{x_i}$ in clause node C_j, then the connection goes from clause node to path P_i and then from path P_j to clause node C_j as shown in Figure 4(c). This completes the construction of instances for the Hamiltonian path.

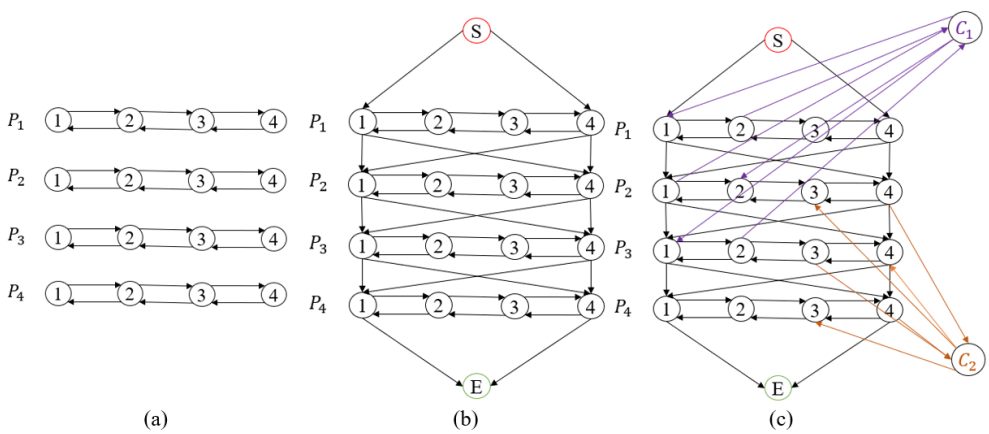

Figure 4: (a) Constructions of paths, (b) Interconnection of paths and adding starting and end node and (c) Instance of the Hamiltonian path

One possible Hamiltonian path is shown in Figure (5), and the truth values of the variables are determined based on the direction in which a path is traversed. If a path P_i is traversed from left to right, $x_i = \mathtt{True}$ or else $x_i = \mathtt{False}$.

Examining the path, we get all the truth values of the variables as True, which is a satisfying assignment for $\phi = (\overline{x_1} \vee x_2 \vee \overline{x_3}) \wedge (\overline{x_2} \vee x_3 \vee \overline{x_4})$. However, we should be very careful because we might have some Hamiltonian path that does not correspond to any valid assignment of the initial formula. Now we provide the reduction from Max3SAT to Max3SAT-7, which is the main new

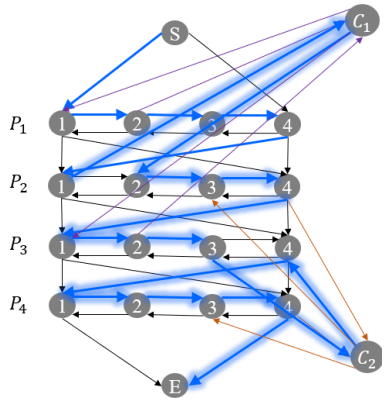

Figure 5: Hamiltonian path from node S to node E

contribution of this project. Some parts of the reduction are very similar to 3-SAT to Hamiltonian path reduction but with minor changes in the gadget creation, highlighting the difference in the current reduction.

We start with a Max3SAT-7 formula ϕ'' with n variables $x_1, x_2, ..., x_n$ and m clauses. We start with creating an unweighted directed graph G. We create n path $P_2, P_2, ..., P_n$ each with 24 nodes, as shown in Figure 6.

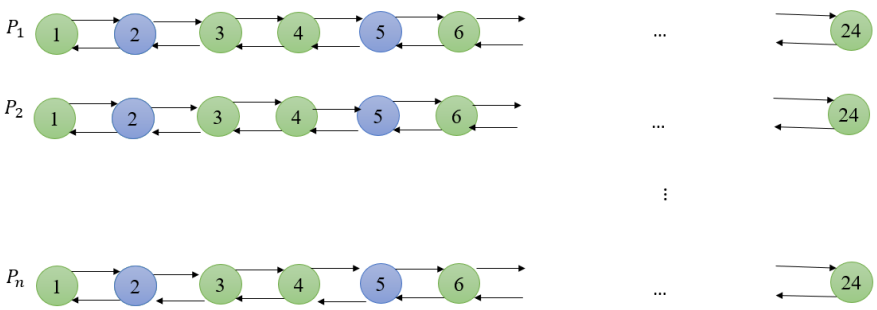

Figure 6: n-Paths each with 24 nodes for n variables.

Each node in a path can have two colors, green and blue. Green nodes form edges with blue nodes or clause nodes, or green nodes of the other paths. Blue nodes form only edges with green nodes of their current path. A blue node is inserted between a pair of green nodes except at the beginning and end of the path (node 1 and node 24). Now we add m clause nodes and interconnect with paths in the same way we did earlier in 3SAT to hamiltonian path reduction, except that connection only goes to the green nodes, as shown in Figure 7.

Finally, we connect the start and end nodes to the path and complete the construction of G. The nodes in the path, including the start and end node, form the subset S of the Max hamiltonian path problem-S, as shown in Figure 8.

This reduction required the proof of a Lemma in order to obtain the L-reduction.

Lemma 2. Any Hamiltonian path passing through all vertices of S is normal.

Proof: we need to define what is a normal path. A path is said to be normal if it corresponds to a valid assignment. This is explained using Figure 9. A normal path traverses each path P_i in only one direction, either from left to right (1 to 24) or from right to left (24 to 1). If suppose

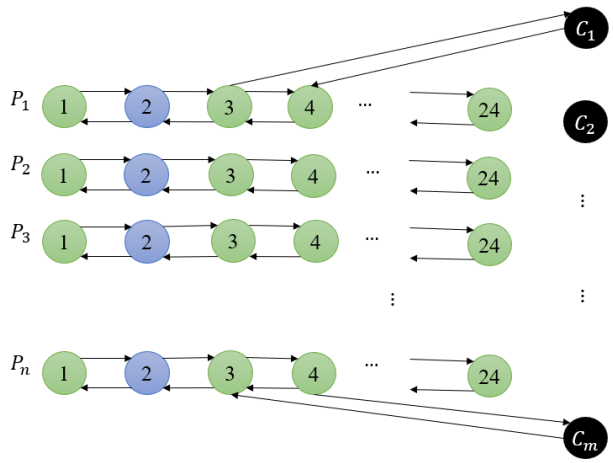

Figure 7: Clause nodes connected to green nodes in each path.

while traversing a path P_1, it goes to visit clause node C_1; it needs to return to the same path P_1 as shown in Figure 9.

Now we use the proof by contraction to show that the lemma is correct. We take a counter-example, as shown in Figure 10.

If suppose while traversing path P_1 after visiting clause node C_1 if we return to node 4 of path P_3, there is no possible way to reach node 4 without visiting any of the nodes in set S twice. Hence, it is not possible for a hamiltonian path to exist if there is no normal path possible through the set S. This proves the lemma. ■

The L-reduction from Max3SAT-7 to Max hamiltonian Path -S implies the following theorem:

Theorem 3. $\mathsf{OPT}(G) = |S| + \mathsf{OPT}(\phi'') = 24n'' + 2 + \mathsf{OPT}(\phi'')$

where $\mathsf{OPT}(G)$ is the maximum possible number of nodes visited in an unweighted directed graph G, and $\mathsf{OPT}(\phi'')$ is the maximum number of clauses satisfied in a Max3SAT-7 formula ϕ'' and n'' is the number of variables.

Proof: We first prove the inequality

$$\mathsf{OPT}(G) \geq |S| + \mathsf{OPT}(\phi'')$$

We have already proved this in the reduction from 3SAT to Hamiltonian Path. Since if we have an optimal assignment of ϕ'', then it corresponds to a Hamiltonian path having $n' = |S| + \mathsf{OPT}(\phi'')$ nodes. Hence, $\mathsf{OPT}(G) \geq |S| + \mathsf{OPT}(\phi'')$.

Now, we prove $\mathsf{OPT}(G) \leq |S| + \mathsf{OPT}(\phi'')$. Let P be an optimal Hamiltonian path passing through S. By the previous lemma, it is normal. So, corresponds to an assignment satisfying m' clauses of ϕ''. Hence, we prove $\mathsf{OPT}(G) \leq |S| + \mathsf{OPT}(\phi'')$. ■

This proves that,

$$\mathsf{Max3SAT\text{-}7} \leq_L \mathsf{MaxHamPathS}$$

Observe that in the previous reduction, we can substitute every clause node with a path of the form P_j with as many nodes as we want. If we do this in such a way that every clause node now is

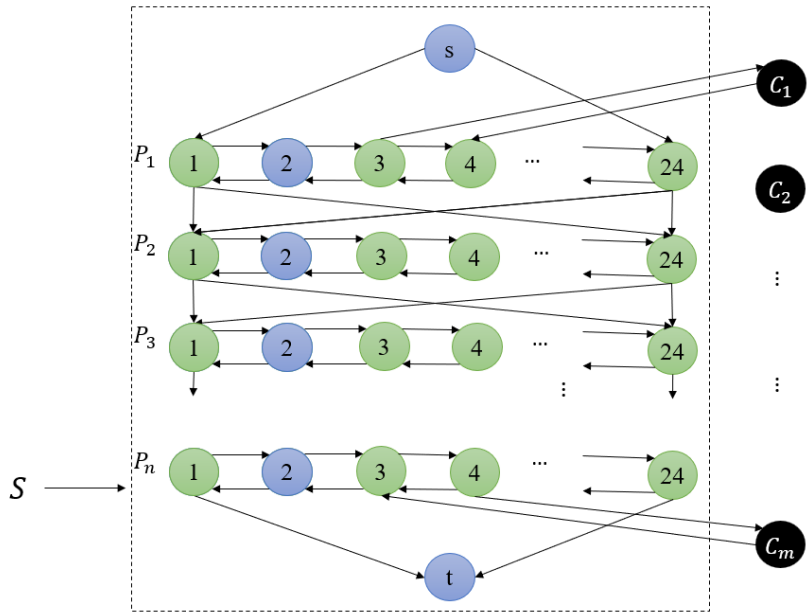

Figure 8: Unweighted directed graph construction showing set S.

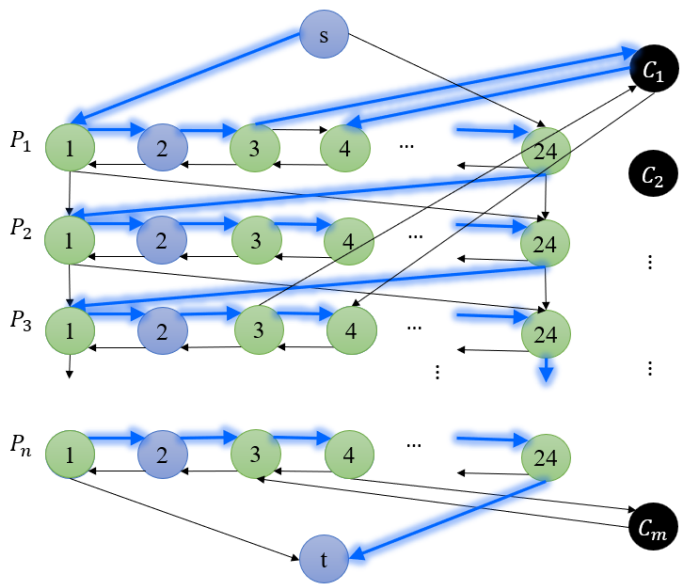

Figure 9: A normal path.

a path with a big number of nodes $B \in \mathbb{N}$, we have the following result:

Theorem 4. Max3SAT-7 $\leq_L$ MaxHamPathS. Moreover, for any fixed $B \in \mathbb{N}$, there is a linear reduction $\phi'' \mapsto (G, S)$ that satisfies

$$\mathtt{OPT}(G) = |S| + \mathtt{OPT}(\phi'')B.$$

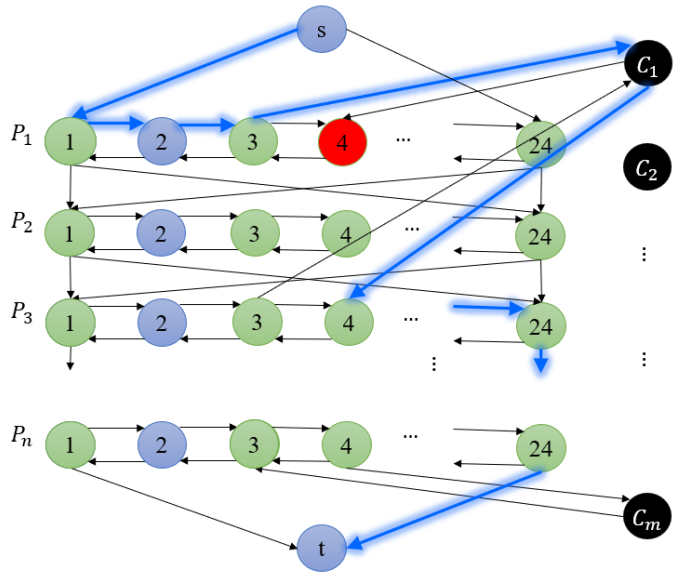

Figure 10: A graph showing a hamiltonian path is not possible if the path is not normal.

Main result

Now we are ready to prove the main result of our project.

Theorem (main). For any $\delta > 0$, if there is a polynomial time algorithm such that for every instance (G, S) of MaxHamPathS, computes a number $\texttt{ALG-OPT}(G)$ such that

$$\texttt{ALG-OPT}(G) \geq (\frac{79}{80} + \delta)\texttt{OPT}(G), \quad \text{then } P = NP.$$

Proof. Let $\delta > 0$. Consider the previous polynomial time reductions $\phi \mapsto \phi' \mapsto \phi'' \mapsto (G, S)$ for some $\epsilon > 0$ and $B \in N$ that we will specify later. Then,

$$\texttt{OPT}(G) = |S| + \texttt{OPT}(\phi'')B = (24n'' + 2) + (\texttt{OPT}(\phi') + 9m')B.$$

Thus, if $\phi \in 3SAT$, then by theorem 1, $\texttt{OPT}(\phi') \geq (1 - \epsilon)m'$, which implies that

$$\texttt{OPT}(G) \geq (24n'' + 2) + (10 - \epsilon)m'B.$$

On the other hand, if $\phi \notin 3SAT$, then $\texttt{OPT}(\phi') \leq (7/8 + \epsilon)m'$, which implies that

$$\texttt{OPT}(G) \leq (24n'' + 2) + (79/8 + \epsilon)m'B.$$

This means that the problem $\text{Gap}(\text{MaxHamPathS}, a, b)$ with

$$a(|G|) = (24n'' + 2) + (10 - \epsilon)m'B, \quad b(|G|) = (24n'' + 2) + (79/8 + \epsilon)m'B$$

is NP-hard. Now, observe that using L'Hôpital's rule,

$$\lim_{B \to +\infty} \frac{b(|G|)}{a(|G|)} = \lim_{B \to +\infty} \frac{(24n'' + 2) + (79/8 + \epsilon)m'B}{(24n'' + 2) + (10 - \epsilon)m'B} = \frac{79/8 + \epsilon}{10 - \epsilon} = \frac{79}{80} + \delta' \quad \text{for } \delta' = \frac{159}{80} \cdot \frac{\epsilon}{10 - \epsilon} > 0$$

and since $n'' = \Theta(m')$, this limit is independent of the values of n'' and m'. This means that we can choose $\epsilon > 0$ sufficiently small, and B sufficiently big (but both fixed) such that for any n', m'', we have that

$$\frac{79}{80} + \delta > \frac{(24n'' + 2) + (79/8 + \epsilon)m'B}{(24n'' + 2) + (10 - \epsilon)m'B} = \frac{b(|G|)}{a(|G|)}.$$

Therefore, for this constant, if there is a polynomial time algorithm such that for every instance (G, S) of MaxHamPathS, computes a number $\texttt{ALG-OPT}(G)$ such that $\texttt{ALG-OPT}(G) \geq (\frac{79}{80} + \delta)\texttt{OPT}(G)$, then

$$\texttt{ALG-OPT}(G) \geq (\frac{79}{80} + \delta)\texttt{OPT}(G) > \frac{b(|G|)}{a(|G|)}\texttt{OPT}(G),$$

and we can conclude that $P = NP$ from lemma 1. ■

References

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